

Implementation and Validation of 1D Return Mapping Algorithms for Plasticity and Viscoplasticity

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Abstract

Computational modeling of inelastic material behavior is essential for predicting structural response under complex loading conditions, particularly for high-temperature applications where creep and rate-dependent effects become significant. Return mapping algorithms provide a robust computational framework for integrating rate-independent and rate-dependent constitutive models within the finite element method. This work presents the implementation and validation of one-dimensional return mapping algorithms for both rate-independent plasticity and viscoplasticity using the Perzyna model.

The algorithms employ an elastic predictor-plastic corrector structure with backward Euler time integration to ensure unconditional stability. For rate-independent plasticity, both perfect plasticity and linear isotropic hardening are implemented. For viscoplasticity, the Perzyna model is used with closed-form solutions for the linear case ($n=1$) and Newton-Raphson iteration for the nonlinear case ($n>1$).

Comprehensive validation studies demonstrate excellent agreement with analytical solutions. For viscoplastic behavior, creep tests show exact agreement with theoretical predictions (creep rate error $< 0.01\%$), stress relaxation exhibits proper exponential decay, and rate sensitivity studies confirm the expected dependence on strain rate. The algorithms provide a solid foundation for extension to multi-dimensional problems and integration with advanced finite element frameworks.

Keywords: Return mapping, Plasticity, Viscoplasticity, Perzyna model, Backward Euler

1. Introduction

Constitutive material models are required to model the material response to loads in finite element analysis. For solid mechanics, variable of interest are the displacement, velocity and acceleration of each and every node to compute derived variables like stresses and strains. For linear behavior, where the stress response follows a linear relationship with strains (by definition displacements) this problem is rather easy to solve but even then, advanced algorithm are used to determine the displacements, stresses and strains efficiently. The same goes for other material behavior like plasticity and viscoplasticity.

Return mapping is a well known algorithm and have been used for well over forty years. Early work by [1] on radial return have paved the way to the work of [2]. Since then, great textbook have also covered the topic and applied it to several material models (constitutive model)[3] [4].

The goal of this mock paper is to condense gathered knowledge about computational plasticity, especially about return mapping applied to 1D viscoplastic constitutive Perzyna model. First, a survey of the theoretical background of return mapping will be presented. Then, the implementation of the algorithms will be shown and finally key results and observation to common material will be studied.

2. Theoretical background

The governing equations in solid mechanics are defined by:

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = \rho \ddot{\mathbf{u}} \quad (1)$$

$$\sigma_{ij} = \mathbb{C}_{ijkl} \varepsilon_{kl} \quad (2)$$

$$\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \quad (3)$$

known as the equilibrium equation (eq. (1)), the constitutive law (eq. (2)) and the strain-displacement relation (eq. (3)).

In a purely linear, 1D isotropic case (2) can be simply seen as the product of the Young's modulus (E) and the strain applied to a rod (ε). In multidimensional nonlinear computational mechanics, this can also be seen as the *trial stress* or *elastic predictor*. As introduced in the last sentence, return mapping starts with an initial guess to see if the new increment of strain generate enough stress so that plasticity or viscoplasticity is induced in the material. To consider that inelastic stress is induced, the *elastic predictor* must give a value that is beyond elastic stress, thus a value larger than σ^y . The stress from a loading is given by a yield function (e.g. Von Mises), and the concept of if the stress state is elastic or inelastic can be verified with the yield criterion ($f(\boldsymbol{\sigma})$). The yield criterion takes at its most simple

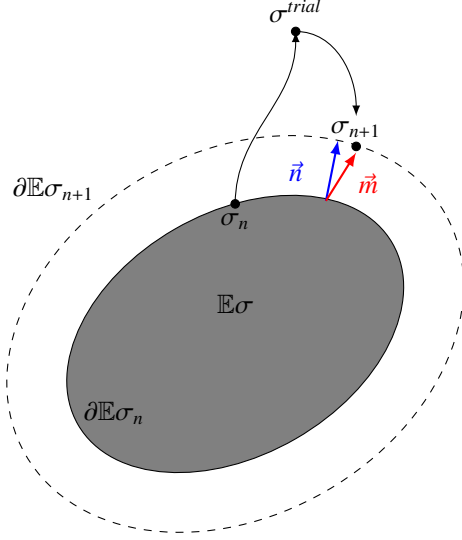


Figure 1: Return mapping in 1D with strain hardening

form, the stress state at the next step ($n + 1$) and the parameter σ^y and is defined as:

$$f(\sigma_{n+1}) = \sigma_{n+1} - \sigma_n^y \quad (4)$$

Then, three conditions has to be met at all times, the Karush-Kuhn-Tucker (KKT) conditions:

$$\gamma \geq 0 \quad (5)$$

$$f(\sigma) \leq 0 \quad (6)$$

$$\gamma f(\sigma) = 0 \quad (7)$$

Here, the symbol γ is called the plastic multiplier and is part of the return mapping algorithm. Essentially, the plastic multiplier scale the vector giving the direction of plastic flow (either $\vec{n}$ or $\vec{m}$) to give the inelastic strain increment (See fig. 1). Here $\vec{n}$ is the vector normal to the yield function and $\vec{m}$ the vector normal to the plastic potential. If $\vec{n} = \vec{m}$, meaning that the yield function and plastic potential are the same, this is called associative plasticity. The opposite case is called anassociative, when $\vec{n} \neq \vec{m}$. Typically, viscoplasticity is associative [4, 3]. This normality condition thus defines the inelastic strain increment:

$$\dot{\epsilon}^{inelastic} = \gamma \mathbf{m} \quad (8)$$

Equations 5, 6, 7 constrain the problem so that for any stress state either $f(\sigma) < 0$ meaning that the stress state is purely elastic or either $\gamma > 0$ meaning inelastic strain is increasing. One other important aspect is that $f(\sigma)$ must at most be equal to zero. As seen in fig. 1, an initial stress state at step (n) is defined by a domain of admissible stress ($\mathbb{E}\sigma$) and a boundary to this domain ($\partial\mathbb{E}\sigma_n$). The way the next stress state is found is, as stated before, when all KKT conditions are met.

2.1. Rate-Independent Plasticity

Rate-independence means that what ever the rate the material is deformed, properties do not change because of it. This however, does not mean that material cannot undergo strain hardening. In this case, it is quite straightforward to obtain the increment of plastic strain. For the sake of simplicity and stay in line with the mock paper objectives, only 1D case will be defined for perfect plasticity ($H = 0$) and isotropic hardening ($H > 0$).

Knowing that the elastic predictor is:

$$\sigma^{trial} = \sigma_n + E\Delta\epsilon \quad (9)$$

we combine it to the extension of eq. (4) to obtain the definition of the next step stress:

$$\sigma_{n+1} = \sigma^{trial} - E\Delta\epsilon^p \quad (10)$$

from eq. (8), and assuming 1D plasticity we then obtain:

$$\sigma_{n+1} = \sigma^{trial} - E\Delta\gamma \text{sign}(\sigma^{trial}) \quad (11)$$

and finally, we rearrange the terms to obtain the increment in plastic multiplier:

$$\Delta\gamma = \frac{f^{trial}}{E} \quad (12)$$

Back to the normality eq. (8), the increment in plastic strain is obtained. Also, similar approach is taken to define the increment of plastic multiplier for plasticity with hardening. Then the increment increment is:

$$\Delta\gamma = \frac{f^{trial}}{E + H} \quad (13)$$

2.2. Viscoplasticity

In some cases and conditions, metals exhibit viscoplastic behavior. Viscoplasticity can be defined by the dependence of material state evolution to the strain rate. This behavior can be represented by a spring-frictional-dashpot element see fig. 2. The dashpot add a relationship with the strain rate and is defined by a viscosity property.

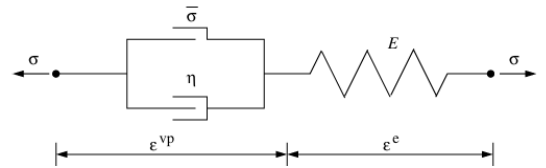


Figure 2: Spring-frictional-dashpot element representation (from [4])

The definition of the viscoplastic strain rate ($\dot{\epsilon}^{vp}$) in the Perzyna model is:

$$\dot{\epsilon}^{vp} = \frac{1}{\eta} \langle \varphi(f) \rangle \mathbf{m} \quad (14)$$

where:

$$\varphi(f) = \left(\frac{f}{\bar{\sigma}} \right)^n \quad (15)$$

The parameters n and η change the behavior of the material:

- n : rate sensitivity exponent
- η : viscosity parameter

Similar development as in section 2.1 and in the relationship between $\dot{\epsilon}^{vp}$ and γ is made. The exception is since it is a time dependent solution, the variables must be integrated. Typically, a backward Euler scheme is used for unconditional stability because it is fully implicit (See [4, 3]). The definition of the increment of the plastic multiplier, for the 1D case is then (in incremental form):

$$\Delta\gamma = \frac{\Delta t}{\eta} \left\langle \frac{f}{\bar{\sigma}} \right\rangle^n \quad (16)$$

For the Perzyna model, the KKT conditions are relaxed, meaning that $f > 0$ is allowed. The reason is that the viscous stress (or over stress) is considered as a complementary state and not included implicitly in the definition in the stress state of the material point since it fades over time. The only limiting case for the Perzyna model is when $\eta \rightarrow 0$ as this solution recovers the rate-independent solution.

The Perzyna model is relatively simple to implement and requires few parameters (minimize the number of tests to calibrate the material model). However, other models and frameworks exist and take into account other mechanics and/or parameters. See appendix Appendix A for an overview of other models.

2.3. Time integration

To integrate time dependant material model, one have to use a time integration scheme. Typically, fully implicit (backward Euler scheme) are used since it is unconditionally stable. For the linear case (where $n = 1$), the integration of the plastic multiplier gives:

$$\Delta\gamma \leftarrow \frac{\Delta t f(\sigma^{trial})}{\eta + \Delta t E} \quad (17)$$

for values of $n > 1$, algorithm such as Newton-Raphson are needed to solve for the increment of the plastic multiplier. This aspect is discussed in the following section (section 3 in algorithm 3)

3. Implementation

In this section, the implementation details of the return mapping algorithm applied to first, the elastoplastic model with linear hardening, and then for the Perzyna-type viscoplastic model in 1D. The extension to higher dimension is pretty straightforward. Note that the implemetenation is shown without material properties. The use of the implementation will be adressd in section 4.

3.1. Algorithm Structure and Code Organization

The algorithm, for either models, have the same basic structure:

- Elastic predictor
- Yield check

- Inelastic corrector

In the case of the linear hardening elastoplastic model, the application is pretty straightforward in regard to the previous list (see algorithm 1). First the algorithm detemine if the stress increment is elastic. If it is not, the next substep is to determine the value of the increment of the plastic multiplier to finally obtain the increment of plastic strain.

Algorithm 1: LINEARHARDENINGRETURNMAPPING Solve the stress state for next stress increment using return mapping for the linear hardening elasto-plastic model

Input: Current step material data $(\sigma_n, \epsilon_n, \epsilon_n^p, \sigma^{y0}, \alpha_n, H)$

Output: Updated material data $(\sigma_{n+1}, \epsilon_{n+1}, \epsilon_{n+1}^p, \alpha_{n+1})$

$\sigma_n^y \leftarrow \sigma^{y0} + H\alpha_n;$

$f(\sigma^{trial}) \leftarrow \sigma_n + E(\Delta\epsilon - \epsilon_n^p) - \sigma_n^y;$

if $f(\sigma^{trial}) < 0$ **then**

$\sigma_{n+1} \leftarrow \sigma^{trial};$

$\epsilon_{n+1} \leftarrow \epsilon_n + \Delta\epsilon;$

$\epsilon_{n+1}^p = \epsilon_n^p;$

else

$\Delta\gamma \leftarrow \frac{f(\sigma^{trial})}{E + H};$

$\Delta\epsilon_p \leftarrow \Delta\gamma \cdot \text{sign}(\sigma^{trial});$

$\epsilon_{n+1}^p \leftarrow \epsilon_n^p + \Delta\epsilon_p;$

$\alpha_n \leftarrow \alpha_n + \Delta\gamma;$

$\sigma_{n+1} \leftarrow \frac{1 - \Delta\gamma E}{|\sigma^{trial}|} \sigma^{trial};$

end

return $\sigma_{n+1}, \epsilon_{n+1}, \epsilon_{n+1}^p, \alpha_{n+1};$

In the case of the Perzyna-type viscoplastic model, the implementation is similar. It only incorporates a nonlinear solve if the strain-rate sensitivity parameter $n > 1$ (See algorithm 2 and algorithm 3) and a temporal integration based on the Backward Euler scheme since it is a time dependant problem.

4. Validation and Results

To evaluate the Implementation introduced in section 3, multiple test regarding plasticity and viscoplasticity have been made. Results regarding those tests are presented in the following sections.

The material used for all tests is a generic metal with the following properties (see table 1):

Table 1: Material properties for the generic metal used in the different test regarding rate independent plasticity and viscoplasticity

Parameter	Value	Units
Young's Modulus (E)	200	GPa
Yield Stress (σ^y)	250	MPa
Hardening modulus (H)	20	GPa
Viscosity (η)	40	GPa*s
Rate exponent (n)	1	—

Algorithm 2: PERZYNARETURNMAPPING Solve the stress state for next stress increment using return mapping for the Perzyna-type viscoplastic model

Input: Current step material data $(\sigma_n, \varepsilon_n, \varepsilon_n^{vp}, \eta, n)$
Output: Updated material data $(\sigma_{n+1}, \varepsilon_{n+1}, \varepsilon_{n+1}^{vp})$
 $f(\sigma^{trial}) \leftarrow \sigma_n + E(\Delta\varepsilon - \varepsilon_n^{vp})$;
if $f(\sigma^{trial}) < 0$ **then**
 $\sigma_{n+1} \leftarrow \sigma^{trial}$;
 $\varepsilon_{n+1} \leftarrow \varepsilon_n + \Delta\varepsilon$;
 $\varepsilon_{n+1}^{vp} = \varepsilon_n^{vp}$;
else
 if $n = 1$ **then**
 $\Delta\gamma \leftarrow \frac{\Delta t f(\sigma^{trial})}{\eta + \Delta t E}$
 else
 $\Delta\gamma \leftarrow \text{NonLinear}\Delta\gamma(f(\sigma^{trial}), \Delta t, E, \sigma^y, \eta, n)$;
 end
 $\varepsilon_{n+1}^{vp} \leftarrow \varepsilon_n^{vp} + \Delta\gamma \cdot \text{sign}(\sigma^{trial})$;
 $\sigma_{n+1} = \sigma^{trial} - E\Delta\gamma \cdot \text{sign}(\sigma^{trial})$
end
return $\sigma_{n+1}, \varepsilon_{n+1}, \varepsilon_{n+1}^{vp}, \alpha_{n+1}$;

Algorithm 3: NonLinear $\Delta\gamma$ Solve the plastic multiplier increment for the Perzyna-type viscoplastic model for the $n > 1$ non-linear case using Newton-Raphson algorithm

Input: $f(\sigma^{trial}), \Delta t, E, \sigma^y, \eta, n$
Output: $\Delta\gamma$
 $itermax \leftarrow 30$;
 $tol \leftarrow 10^{-10}$;
 $\Delta\gamma \leftarrow \frac{\frac{\Delta t}{\eta} \left(\frac{f_{trial}}{\sigma^y} \right)^n}{1 + n \frac{\Delta t E}{\eta (\sigma^y)^n}}$;
 $k \leftarrow 0$;
while $k < itermax$ **do**
 $f_{current} \leftarrow f_{trial} - E\Delta\gamma$;
 $R = \Delta\gamma - \frac{\Delta t}{\eta} \left(\frac{f_{trial}}{\sigma^y} \right)^n$;
 $\frac{\partial R}{\partial \Delta\gamma} \leftarrow 1 + n \frac{\Delta t E}{\eta (\sigma^y)^n} \frac{f_{current}}{\sigma^y}^{n-1}$;
 $\Delta\gamma \leftarrow \Delta\gamma - \frac{R}{\frac{\partial R}{\partial \Delta\gamma}}$
end
return $\Delta\gamma$;

All test are equivalent of solving the equation for one quadrature point in a finite element model. i.e. The equations solved do not rely on a finite element. Also, generic parameters will be changed during certain tests to evaluate the sensitivity to this parameter.

4.1. Rate-Independent Plasticity

The tests consist of loading the material up until $\varepsilon = 0.02$, then unload and reload to $\varepsilon = 0.03$. Tests will be the same for perfect plasticity and isotropic hardening.

4.1.1. Perfect Plasticity

In the case of perfect plasticity, as expected the stress plateaus at the yield value (see fig. 3). The accumulation of plastic strain makes the unloading down to $\varepsilon = 0$ generate compressive stress, hence the negative stress. On the reload, the stress plateaus again up until the final strain target of $\varepsilon = 0.03$

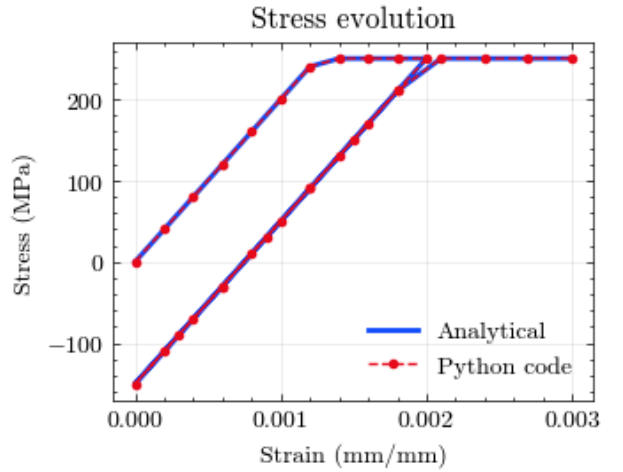


Figure 3: Evolution of stress using a perfect plasticity model, with one unloading

It can be seen (fig. 4), that the plastic strain starts at 0 and plateaus at a value of approximately $\varepsilon^p = 0.00075$ during unloading. Then rise up again during the final loading.

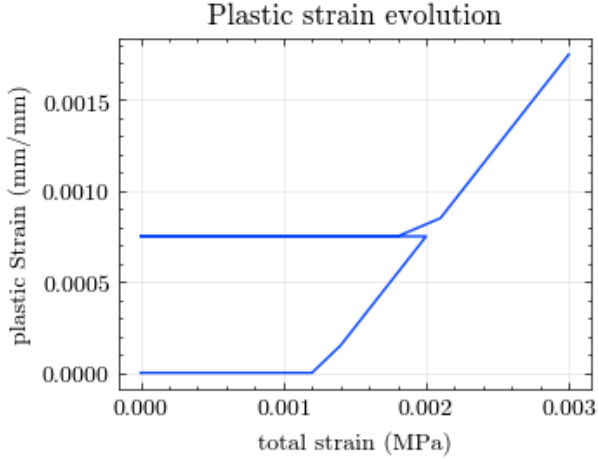


Figure 4: Evolution of plastic strain using a perfect plasticity model, with one unloading

4.1.2. Isotropic Hardening

In the case of isotropic hardening, the load path is the same. However, the hardening parameter H is varied to study the effect on the resulting stress in the material. As seen in fig. 5, the perfectly plastic case ($H = 0$) still follows the same intended behavior: plateaus at yield stress. Then the H parameter is changed from $H = 1000 \text{ MPa}$ to $H = 50\,000 \text{ MPa}$. The higher the value, the greater the slope during hardening.

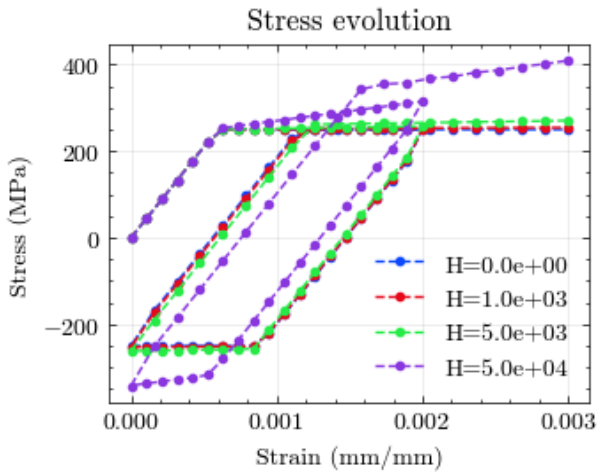


Figure 5: Evolution of stress using an isotropic plasticity model, with one unloading and multiple hardening parameters

4.2. Viscoplasticity

For rate dependent case (viscoplastic), tests have to be different to evaluate the rate effects on material properties.

4.2.1. Rate Sensitivity

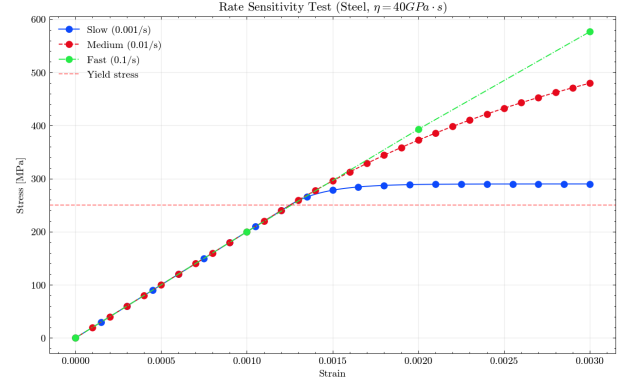


Figure 6: Effect of rate sensitivity on material response with Perzyna type viscoplastic model

4.2.2. Creep Test

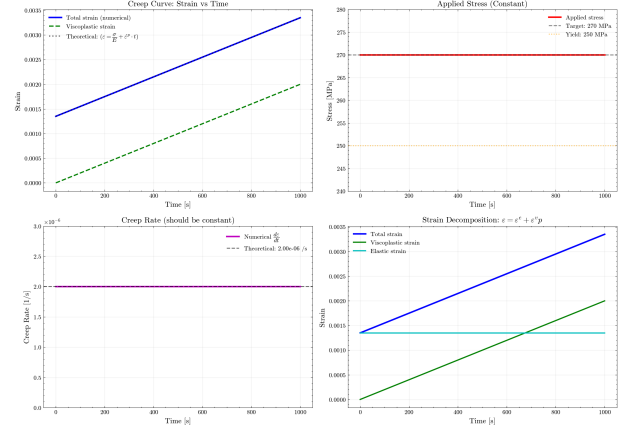


Figure 7: Stress response from creep test with Perzyna type viscoplastic model

4.2.3. Stress Relaxation

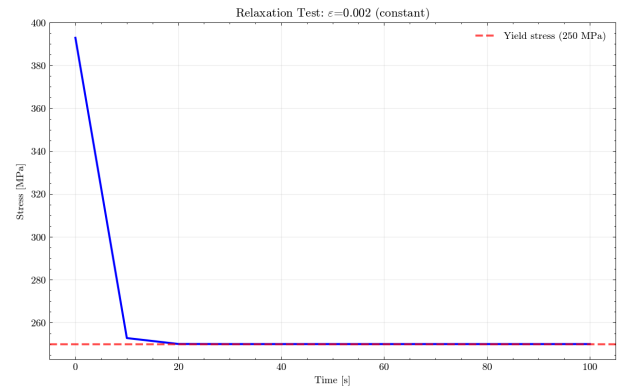


Figure 8: Stress response from relaxation test with Perzyna type viscoplastic model

4.2.4. Viscosity Effect

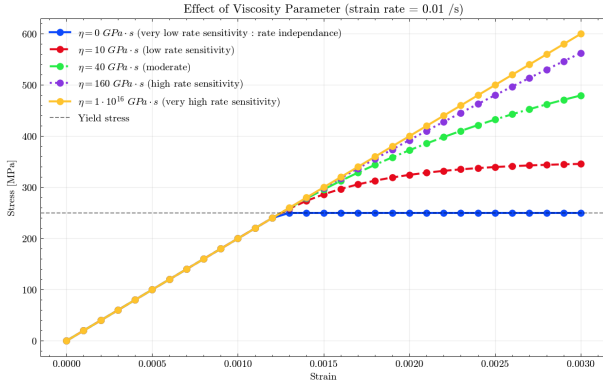


Figure 9: Viscosity sensibility test with Perzyna type viscoplastic model

5. Discussion

6. Conclusion and Future Works

Appendix A. Viscoplastic models and frameworks

In this appendix, some models and extension of the Peryzyna-type model are shown. Not intended to be a thorough instruction, the following information should be seen as an overview and insight.

Here, parameters are identified through the mean of optimization, identification algorithm or any other numerical method (e.g. Levenberg-Marquardt algorithm, genetic algorithm, machine learning, etc.). To name a few:

- Duvaut-Lions: add a back-bone stress concept ($\bar{\sigma}$). Viscoplastic strain is then computed as:

$$\dot{\epsilon}^{vp} = \frac{1}{\tau} (\mathbf{D}^e)^{-1} 1(\sigma - \bar{\sigma}) \quad (\text{A.1})$$

where τ is the relaxation time and $\mathbf{D}^e$ the elastic stiffness matrix linked to material properties. Back-bone stress is the stress state in the rate-independent case. Meaning that on every iteration, the rate-independent case has to be solved along the viscoplastic state.

- Consistency model: The stress state must lie on the yield surface. Thus the redefinition of eq. (6) gives:

$$f(\sigma, \dot{\epsilon}^{vp}) \leq 0 \quad (\text{A.2})$$

- Johnson-Cook [5]: An empirical model, meaning it is based on paramete calibration rather than material thermodynamics, with the form:

$$\sigma_y = \underbrace{(A + B\epsilon_p^n)}_{\text{strain hardening}} \cdot \underbrace{(1 + C \ln \dot{\epsilon}^*)}_{\text{strain rate sensitivity}} \cdot \underbrace{(1 - T^{*m})}_{\text{thermal softening}} \quad (\text{A.3})$$

The advantages of the Johnson-cook model is that it is the yield stress that is defined through the model, therefore the stress can be evaluated with any other yield function. An

other advantage is the modularity. If for example the material do not exhibit strain hardening or thermal softening, the equation becomes:

$$\sigma_y = A + C \ln \dot{\epsilon}^* \quad (\text{A.4})$$

- Chaboche [6]: Framework used in cyclic viscoplasticity. It includes:
 - Isotropic hardening
 - Multiple backstress tensors with Armstrong-Frederick kinematic hardening rules
 - Viscous flow rule (Typically Norton-type or Perzyna-type)

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